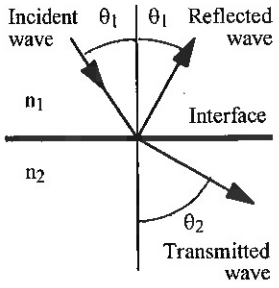
Optoelectronics Solutions 2019

1. a) Snell's law $n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$ relates the angles of incidence and refraction θ_1 and θ_2 when light strikes an interface between media with refractive indices n_1 and n_2 as shown below.



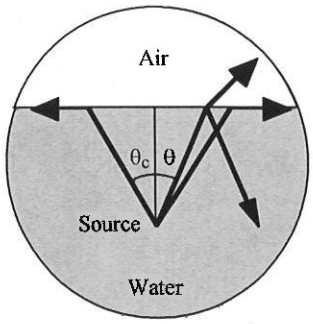
[3]

Re-arranging $n_2 = n_1 \sin(\theta_1) / \sin(\theta_2)$

Assuming that $\theta_1 = 30°$, $\theta_2 = 22°$ and $n_1 = 1$ (air), we get $n_2 = \sin(30) / \sin(22) = 1.33$.

[1]

b) The figure below shows a light source immersed in water. Light from the source will be reflected and refracted at the interface between water and air. When θ reaches the critical angle θ_C, reflection will be total. Thus, only light contained in a cone of half-angle θ_C can escape.



[4]

c) At the critical angle, $\theta_2 = 90°$, so $\theta_1 = \sin^{-1}(n_2/n_1) = \theta_C$.

For the semiconducting slab, the critical angle is $\theta_C = \sin^{-1}(1/3.5) = 16.6°$.

[2]

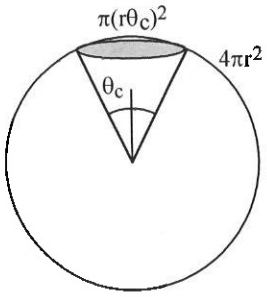
Similarly, the amplitude reflection coefficient at normal incidence is $\Gamma = (n_2 - n_1)/(n_2 + n_1)$

For the semiconducting slab and air, $\Gamma = (3.5 - 1)/(3.5 + 1) = 0.555$

[2]

The critical angle is quite small in this case, and the reflection coefficient quite large

d) Since the critical angle is small, we can use small angle approximations to estimate the fraction of power that can escape. Since $\theta_C = \sin^{-1}(1/n_1)$, and n_1 is large, $\theta_C \approx 1/n_1$. From the diagram below, the fraction of power in a cone of half-angle θ_C is $\pi(r\theta_C)^2 / 4\pi r^2 = \theta_C^2/4 = 1/4n_1^2$.



[3]

Since the reflection coefficient is large, we must also take reflection into account.

The power transmission coefficient at normal incidence is $T^2 = 1 - \Gamma^2$, or:

$T^2 = 1 - (n_2 - n_1)^2/(n_2 + n_1)^2 = \{(n_2 + n_1)^2 - (n_2 - n_1)^2\} / (n_2 + n_1)^2$, or:

$T^2 = 4n_1n_2/(n_2 + n_1)^2$

Since $n_2 = 1$, $T^2 = 4n_1/(n_1 + 1)^2$

The fraction of power leaving the surface is then $\eta = T^2 \times 1/4n_1^2 = 1/\{n_1(n_1 + 1)^2\}$

[2]

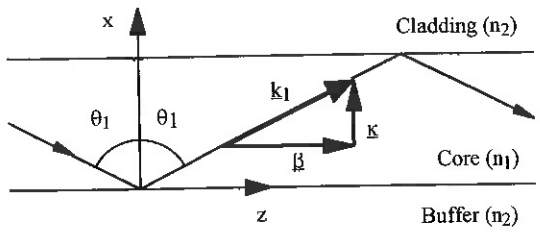
In this case, the external efficiency is $\eta = 1/(3.5 \times 4.5^2) = 0.0141$

[1]

e) The value of 1.41% above is so low that LEDs will be extremely inefficient sources in communication systems. However, since the refractive index of water is so much closer to unity, more light will escape and we can expect underwater swimming pool lights to work well, however they operate.

[2]

2a) The structure of a symmetric slab dielectric waveguide is as shown below.



[2]

In a zig-zag picture of ray propagation in the core, rays are continually incident upon and reflected from interfaces at an angle θ_1. The inclined vector $\underline{k}_1$ (whose magnitude is $k_0 n_1$) is the local propagation vector. Its horizontal component $\underline{\beta}$ whose magnitude is $\beta = k_0 n_1 \sin(\theta_1)$ represents the component of propagation along the guide. The vertical component $\underline{\kappa}$ whose magnitude is $\kappa = \sqrt{(n_1^2 k_0^2 - \beta^2)}$ represents the component of propagation perpendicular to the guide.

[3]

b) Snell's law is $n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$. The onset of total internal reflection (the cutoff condition) occurs when $\theta_2 = \pi/2$.

[2]

Under these conditions, $n_1 \sin(\theta_1) = n_2$ so the cutoff value of β is $k_0 n_2$.

Consequently, the cutoff value of $\kappa = \sqrt{(n_1^2 k_0^2 - \beta^2)}$ is $k_0 \sqrt{(n_1^2 - n_2^2)}$.

Similarly, the cutoff value of $\gamma = \sqrt{(\beta^2 - n_2^2 k_0^2)}$ is zero.

[3]

c) From the above, the eigenvalue equations:

$\tan(\kappa h/2) = \gamma/\kappa$

$\tan(\kappa h/2) = -\kappa/\gamma$

reduce at cutoff to:

$\tan(\kappa h/2) = 0$ so $\kappa h/2 = 0, \pi, 2\pi \ldots$

$\tan(\kappa h/2) = -\infty$ so $\kappa h/2 = \pi/2, 3\pi/2 \ldots$

The general cutoff condition is therefore $\kappa h/2 = \nu\pi/2$, or:

$(k_0 h/2) \sqrt{(n_1^2 - n_2^2)} = \nu\pi/2$ with $\nu = 0, 1, 2 \ldots$

Since $k_0 = 2\pi/\lambda$, this can be rewritten as $(2h/\lambda) \sqrt{(n_1^2 - n_2^2)} = \nu$

[4]

If a guide is to be single moded, all modes except the lowest order mode ($v = 0$) must be cut off.

The condition for single-mode operation is therefore $(2h/\lambda)\sqrt{(n_1^2 - n_2^2)} \leq 1$

[2]

d) Re-arranging the above, we can write the condition on guide thickness as $h \leq \lambda/\{2\sqrt{(n_1^2 - n_2^2)}\}$

If $n_1 = 1.51$, $n_2 = 1.5$ and $\lambda = 1.5$ μm, we obtain $h \leq 4.33$ μm.

[2]

Rearranging further, we get:

$\sqrt{(n_1^2 - n_2^2)} \leq (\lambda/2h)$, so

$n_1^2 \leq n_2^2 + (\lambda/2h)^2$

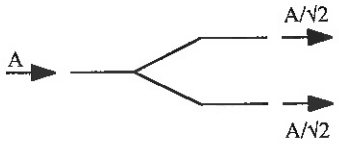
A core thickness of $h = 6$ μm can therefore be achieved if:

$n_1^2 \leq 1.5^2 + (1.5/12)^2 = 2.2656$ so the core index needed is $n_1 \leq 1.505$

[2]

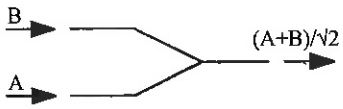
3. a) Assuming perfect symmetry:

i) The structure below will act as an equal, lossless power divider. If the input amplitude is A, the input power is proportional to A^2. The two equal output powers are then both proportional to $A^2/2$, so the output amplitudes must both be $A/\sqrt{2}$. The scaling factor of $\sqrt{2}$ should be noted, since it occurs repeatedly in later parts.



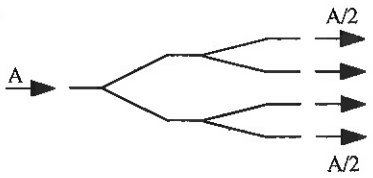
[2]

ii) The structure below is outwardly similar, but must be tackled more carefully. Particularly, it supports two modes on the left-hand side of the fork (symmetric and anti-symmetric), not one. The input amplitude distribution must therefore be resolved into symmetric and anti-symmetric components before analysis. The former consists of amplitudes (A + B)/2 in each guide, and the latter of amplitudes (A - B)/2 in the lower guide and (B - A)/2 in the upper guide. At the fork, the anti-symmetric mode will be radiated, so this component may be ignored. The power contained in the symmetric component will emerge from the guided port; this component is proportional to $2 \times (A + B)^2/4 = (A + B)^2/2$. The output amplitude is then $(A + B)/\sqrt{2}$.



[2]

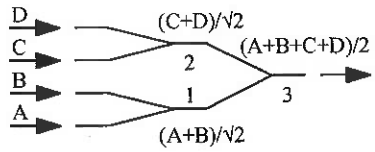
iii) The structure below may be analysed by repeating the steps in part i). Alternatively, it may be noted that the output powers must all be proportional to $A^2/4$, leading to the direct deduction that each output must have amplitude A/2.



[3]

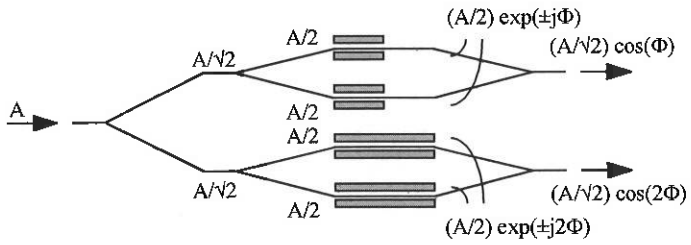
iv) The structure below may be analysed by noting that only the symmetric mode amplitude (A + B)/2 will contribute to the output from the lower left Y-junction at point 1, as $(A + B)/\sqrt{2}$.

Similarly, only the corresponding amplitude (C + D)/2 will contribute to the output from the upper left Y-junction at point 2, as (C + D)/√2. These two components may again be resolved into symmetric and anti-symmetric components. Since the former has amplitude (A + B + C + D)/2√2, the output amplitude at point 3 is then (A + B + C + D)/2.



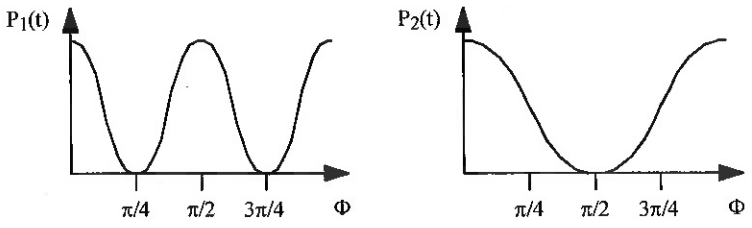
[3]

b) After the first and second sets of Y-junctions, the amplitudes are A/√2 and A/2 as below. Ignoring common mode phase shifts, the amplitudes before the third Y-junctions must be (A/2) exp(±j2Φ) and (A/2) exp(±jΦ). These amplitudes can be resolved into symmetric and anti-symmetric components. The symmetric modes have amplitude (A/2) cos(2Φ) and (A/2) cos(Φ). Consequently, the output amplitudes are $A_1 = (A/\sqrt{2})\cos(\Phi)$ and $A_2 = (A/\sqrt{2})\cos(2\Phi)$.



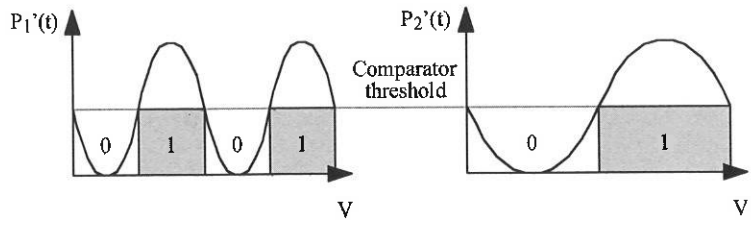
[4]

The output powers must vary as $P_1 = (P/2)\cos^2(2\Phi)$ and $P_2 = (P/2)\cos^2(\Phi)$, where P is the launched power. These powers vary with Φ as shown below.



[2]

c) The responses above may be shifted to the new variations P_1' and P_2' as below, using static phase biases. If the signals are then detected and passed through comparators whose threshold is set to half the maximum, the output is a 2-bit representation of the drive, i.e. a flash A-D encoding.

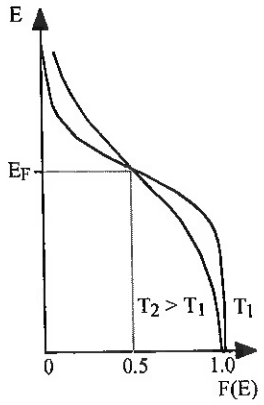


[4]

4.a) F(E) dE represents the probability of occupation of energy levels between E and E + dE. E_F is the energy level for which F(E) = 0.5, and hence represents the equilibrium level.

[2]

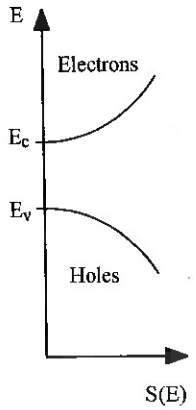
The Fermi-Dirac function $F(E) = 1/\{1 + \exp[(E - E_F)/KT]\}$ varies with energy at different temperatures as shown below:



[2]

b) For electrons, the density of states function $S(E) = C_e \sqrt{(E - E_C)}$ implies the parabolic relation $E = E_C + (S/C_e)^2$ between E and S for $E \geq E_C$. Similarly, for holes, the density of states function $S(E) = C_h \sqrt{(E_V - E)}$) implies the parabolic relation $E = E_V - (S/C_h)^2$ between E and S for $E \leq E_V$. Consequently, the two variations are as shown below.

[2]



[2]

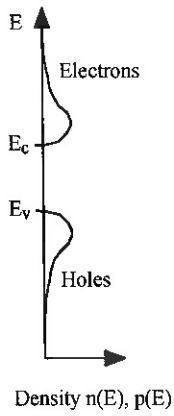
c) For $E >> E_F$, $F(E) \approx 1/\exp[(E - E_F)/KT] = \exp[(E_F - E)/KT]$

$1 - F(E) = \exp[(E - E_F)/KT] /\{1 + \exp[(E - E_F)/KT]\}$

For $E << E_F$, $1 - F(E) \approx \exp[(E - E_F)/KT]$

[2]

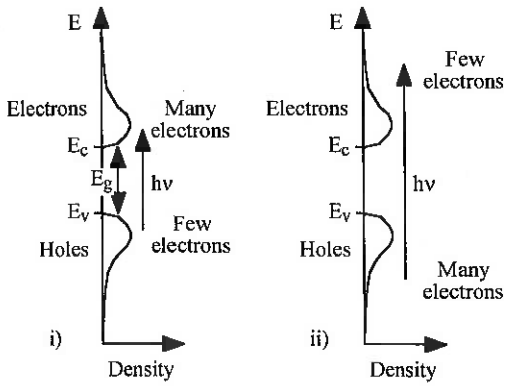
The electron distribution n(E) can be found by multiplying S(E) by F(E).
Consequently, n(E) can be found by multiplying S(E) by an exponential that decays as E rises.
The electron distribution p(E) can be found by multiplying S(E) by 1 - F(E).
Consequently, p(E) can be found by multiplying S(E) by an exponential that decays as E falls.



[4]

d) For photon energies hν close to $E_g = E_C - E_V$, optical absorption requires the promotion of electrons from states containing many holes (and consequently few electrons) to states containing many electrons, as shown in i) below. Since the Pauli exclusion principle allows the occupation of each state by only two electrons, the probability of such a transition occurring is low. Conversely, for $h\nu >> E_g$, absorption requires the promotion of electrons from states containing few holes (and hence many electrons) to states containing few electrons, as shown in ii) below. The probability of such a transition is much higher, so absorption is also higher.

[3]



[3]

5. a) In a laser, one electron will ideally combine with one hole to create one photon. Quantum efficiency describes the reduced number of photons per electron that is achieved in practice. Similarly, in a photodiode, one photon will ideally generate one electron-hole pair. Quantum efficiency describes the reduced number of electrons per photon that is achieved in practice.

[2]

The main factors limiting quantum efficiency in LEDs and lasers are i) diffusion of carriers away from the active region, and ii) non-radiative recombination.

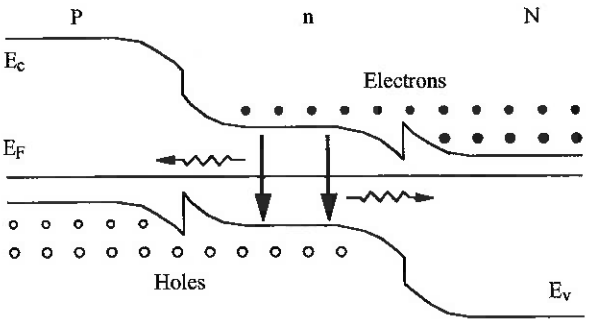
[2]

The main factors limiting quantum efficiency in photodiodes are recombination of electron-hole pairs before they reach the contacts.

[1]

b) Diffusion of injected carriers away from the active region can be prevented in a laser using a double heterostructure, which provides separate energy barriers for electrons and holes on either side of the active region. For a given material, there is nothing that can be done to prevent non-radiative recombination.

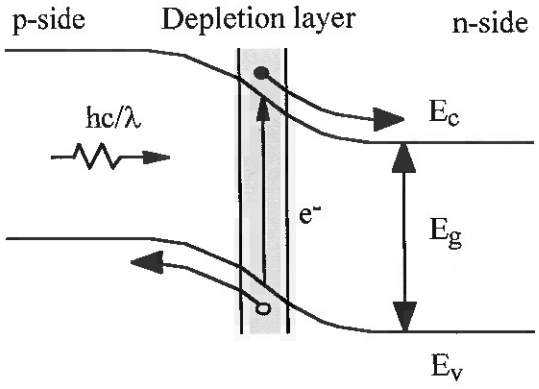
[3]



[2]

c) Recombination of photogenerated carriers is primarily prevented by absorption in the depletion layer of a diode structure, whose strong electric field separates the carrier pairs before they can recombine. Photons can be prevented from passing through the depletion layer without absorption using a p-i-n layer structure, which increases the depletion layer thickness. Photons can be prevented from being absorbed before they reach the depletion layer using a material with a wide bandgap as a surface layer, in a heterostructures such as a P-i-n photodiode.

[3]



[2]

d) Assuming that h is Planck's constant, c is the speed of light, λ is the wavelength, e is the electron charge and E_g is the bandgap in electron volts:

In a beam of optical power P, there are Pλ/hc photons/sec at wavelength λ.

If each photon generates an electron-hole pair, the photocurrent is $I_{pideal} = Pe\lambda/hc$ A.

The ideal responsivity $R_{ideal} = I_{pideal}/P$ is then $R_{ideal} = e\lambda/hc$ A/W

Taking into account the quantum efficiency η, this modifies to $R_{actual} = \eta e\lambda/hc$ A/W

However, absorption cannot occur unless the $hc/\lambda > eE_g$, i.e. up to a wavelength $\lambda_g = hc/eE_g$.

[3]

Re-arranging the above, $\eta = R_{actual}hc/e\lambda$. Combination with the data gives the table below.

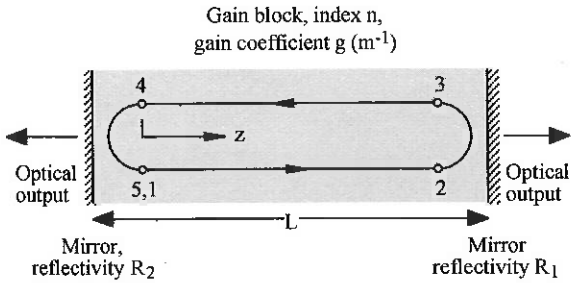
Since the quantum efficiency at λ = 0.9 μm exceeds unity, this data must be corrupted.

Furthermore, if E_g =1.12 eV, $\lambda_g = 6.62 \times 10^{-34} \times 3 \times 10^8 / (1.6 \times 10^{-19} \times 1.12) = 1.108 \times 10^{-6}$ m, or 1.11 μm. Consequently, the entry for λ = 1.2 μm must also be corrupted. Other entries are realistic.

Wavelength (μm)	Responsivity (A/W)	Quantum efficiency
0.7	0.378	0.67
0.8	0.483	0.75
0.9	**0.761**	**1.05**
1.0	0.322	0.40
1.2	**0.193**	**0.20**

[2]

6. a) The figure below shows a suitable model for the calculation. The laser has effective index n (so the propagation constant is $\beta = 2\pi n/\lambda$ at wavelength λ), gain g and length L. The end mirrors have reflectivity R_1 and R_2.



Assuming the electric field amplitude is $E_1 = E_0$ at point 1, the values at points 2 - 5 are:

Point 2: $E_2 = E_0 \exp(-j\beta L) \exp(+gL)$

Point 3: $E_3 = E_0 \exp(-j\beta L) \exp(+gL)\, R_1$

Point 4: $E_4 = E_0 \exp(-j2\beta L) \exp(+2gL)\, R_1$

Point 5: $E_5 = E_0 \exp(-j2\beta L) \exp(+2gL)\, R_1R_2$

For oscillation, we must have $E_1 = E_5$.

Consequently, the overall oscillation condition is $\exp(-j2\beta L) \exp(+2gL)\, R_1R_2 = 1$

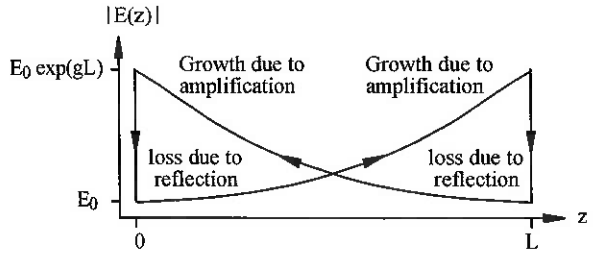
[3]

Clearly, the LHS above must be real. Hence the phase condition is $\exp(-j2\beta L) = 1$ so $\beta L = \nu\pi$.

The gain condition must then be $\exp(+2gL)\, R_1R_2 = 1$

[2]

The field amplitude grows exponentially in either direction inside the cavity, but is reduced by reflection at either end so that the overall variation forms a closed curve as shown below.



[2]

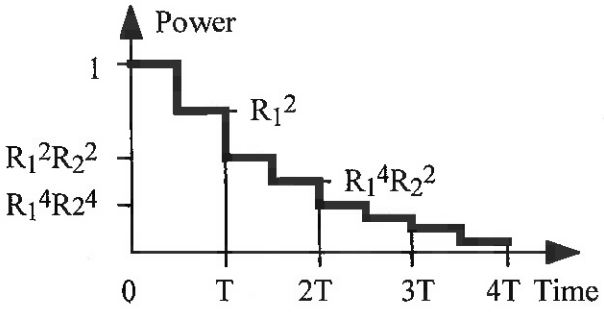
b) Each cavity transit takes a time $T = 2L/v_g$, where $v_g \approx c/n$ is the group velocity.

Ignoring the gain:

The remaining power after 1 transit is $(R_1R_2)^2$

The remaining power after N transits is $(R_1R_2)^{2N}$

The power therefore decays in a series of discrete steps as shown below.



[2]

The number of transits required to reduce the power to 1/e can be found from

$(R_1R_2)^{2N} = 1/e$ so $N = 1/\{2 \log_e(1/R_1R_2)\}$

These transits take a time $\tau_P = NT$

The photon lifetime is then $\tau_P = L/\{v_g \log_e(1/R_1R_2)\}$

[3]

c) The phase condition can be written as $(2\pi n/\lambda)L = \nu\pi$. Hence, $\lambda = 2nL/\nu$, where ν is integer.

The separation between modes is therefore $\Delta\lambda = 2nL/\nu - 2nL/(\nu + 1) = 2nL/\{\nu(\nu + 1)\}$

If n is reasonably large, this result can be approximated as $\Delta\lambda = 2nL/\nu^2$, or $\Delta\lambda = \lambda^2/2nL$

If $\lambda = 1.5 \times 10^{-6}$ m, n = 3.5 and $L = 250 \times 10^{-6}$ m:

$\Delta\lambda = (1.5 \times 10^{-6})^2/(2 \times 3.5 \times 250 \times 10^{-6}) = 1.286 \times 10^{-9}$ m, or 1.286 nm.

[3]

The gain condition can be written as $g = (1/2L) \log_e(1/R_1R_2)$

Assuming the mirrors are formed from the interface between the laser material and air, the reflectivity is $R_1 = R_2 = (n - 1)/(n + 1) = (3.5 - 1)/(3.5 + 1) = 0.555$

The gain needed is then $g = \log_e(1/0.555^2) / (2 \times 250 \times 10^{-6}) = 2355$ m^{-1}.

[3]

For the laser here, $\tau_P = 250 \times 10^{-6}/\{(3 \times 10^8/3.5) \log_e(1/0.555^2)\} = 2.48 \times 10^{-12}$ s, or 2.48 ps.

[2]